

EXAMPLE

We toss a coin n times, indep.
Head w/ prob p , tails w/ prob $1-p$.
What is the expected # streaks?

Solution length ≥ 1

Let X = #streaks in n tosses

Let X_i = indicator of a new streak starting w/ i th toss

$$= \begin{cases} 1, & \text{new streak on toss } i \\ 0, & \text{o/w} \end{cases}$$

Then, $X = \sum_{i=1}^n X_i \Rightarrow$ by lin of exp, $E(X) = \sum_{i=1}^n E(X_i) = \sum_{i=1}^n P(\text{new streak at toss } i)$ P_i

Notice, $P_1 = 1$

$$\text{For } i \geq 2, P_i = P(\text{toss } i \neq \text{toss } i-1) = P(\{\text{toss } i = T, \text{toss } i-1 = H\} \cup \{\text{toss } i = H, \text{toss } i-1 = T\})$$

$$\begin{aligned} & \xrightarrow{\text{indep. + additivity}} P(\text{toss } i = T)P(\text{toss } i-1 = H) + P(\text{toss } i = H)P(\text{toss } i-1 = T) \\ &= p(1-p) + (1-p)p \\ &= 2p(1-p) \end{aligned}$$

$$\therefore E(X) = \sum_{i=1}^n P_i = 1 + \sum_{i=2}^n 2p(1-p) = 1 + 2p(1-p)(n-1)$$

EXAMPLE

Quicksort (my TCS ass is happy for once)

① Pick a number unif rand

② Compare the other numbers to the pivot and sort them as "smaller" or "bigger" parts accordingly

③ Sort sublists accordingly

④ Return "smaller" pivot "bigger"

How many times do we need to compare the numbers?

Solution

Let X = #comparisons, compute $E(X)$ for ordering $x_1, x_2, \dots, x_{n-1}, x_n$

For $1 \leq i < j \leq n$, let

$$X_{i,j} = \begin{cases} 1, & x_i \text{ and } x_j \text{ are compared} \\ 0, & \text{o/w} \end{cases}$$

Then, $X = \sum_{1 \leq i < j \leq n} X_{i,j}$. By lin of exp, $E(X) = \sum_{1 \leq i < j \leq n} E(X_{i,j})$

WLOG, relabel s.t. $x_1 < x_2 < \dots < x_i < x_{i+1} < \dots < x_{j-1} < x_j < \dots < x_n$

Consider the interval in which we choose a pivot in $\{x_i, x_{i+1}, \dots, x_{j-1}, x_j\}$ (o/w they won't be compared)

Then, x_i, x_j are compared $\Leftrightarrow$ one of them is the pivot

$$\therefore P(x_i, x_j \text{ compared}) = \frac{2}{j-i+1}$$

$$\therefore E(X) = \sum_{i,j} P(x_i, x_j \text{ compared}) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{2}{j-i+1} \overset{j=i+1}{=} \sum_{i=1}^{n-1} \sum_{j=1}^{n-i} \frac{2}{j+1} \approx 2 \sum_{i=1}^{n-1} \ln(n-i) \approx 2n \ln n$$

MOMENTS OF NUMBER OF EVENTS

Let $A_1, A_2, \dots, A_n$ be events in a probability space (no need to be indep!)

Let X = #events that occur

$$\Rightarrow X = \sum_{i=1}^n I_i, I_i = \begin{cases} 1, & A_i \text{ occurs} \\ 0, & A_i \text{ doesn't occur} \end{cases} \Rightarrow E(X) = \sum_{i=1}^n E(I_i) = \sum_{i=1}^n P(A_i)$$

What about the variance?

$$\binom{n}{2} = \# \text{pairs of events that both occur} = \sum_{1 \leq i < j \leq n} \mathbb{I}_{\{i, j\}}$$

$$\therefore \mathbb{E}\left[\binom{n}{2}\right] = \mathbb{E}\left[\sum_{1 \leq i < j \leq n} \mathbb{I}_{\{i, j\}}\right] \stackrel{\text{lin of exp}}{=} \sum_{1 \leq i < j \leq n} \mathbb{E}[\mathbb{I}_{\{i, j\}}] \stackrel{\text{lin of exp}}{=} \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j)$$

$$\text{As } \binom{n}{2} = \frac{x(x-1)}{2}, \text{ thus } \mathbb{E}\left[\binom{n}{2}\right] = \mathbb{E}\left[\frac{x^2 - x}{2}\right] = \frac{1}{2}\mathbb{E}[x^2] - \frac{1}{2}\mathbb{E}[x]$$

$$\therefore \text{Var}(x) = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = 2\mathbb{E}\left[\binom{n}{2}\right] + \mathbb{E}[x] - \mathbb{E}[x]^2$$

EXAMPLE

$X \sim \text{Bin}(n, p)$

X counts #succ in n indep trials, w/ succ prob p .

Let $A_i = \{i\text{th trial succ}\}$

$$\hookrightarrow \mathbb{E}[X] = \sum_{i=1}^n \mathbb{P}(A_i) = \sum_{i=1}^n p = np$$

$$\hookrightarrow \mathbb{E}\left[\binom{n}{2}\right] = \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) \stackrel{\text{indep}}{=} \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i)\mathbb{P}(A_j) = \sum_{1 \leq i < j \leq n} p^2 = \binom{n}{2} p^2$$

$$\therefore \text{Var}(X) = 2\mathbb{E}\left[\binom{n}{2}\right] + \mathbb{E}[X] - \mathbb{E}[X]^2 = 2\binom{n}{2} p^2 + np - n^2 p^2 = np(1-p)$$

EXAMPLE

Hat return problem

n people, n hats, hats returned unif at random

$X = \#$ people getting their own hats

$A_i = \{i\text{th person gets own hat}\}$, where $p_i = \mathbb{P}(A_i) = \frac{1}{n}$ (notice, A_i, A_j not indep)

$$\text{Then, } \mathbb{E}[X] = \sum_{i=1}^n p_i = n\left(\frac{1}{n}\right) = 1 \quad \text{by def}$$

$$\mathbb{E}\left[\binom{n}{2}\right] = \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) \stackrel{\text{by def}}{=} \sum_{i < j} \mathbb{P}(A_i) \mathbb{P}(A_j | A_i) = \sum_{i < j} \frac{1}{n} \left(\frac{n-1}{n}\right) = \binom{n}{2} \left(\frac{n-1}{n}\right) = \frac{1}{2}$$

$$\therefore \text{Var}(X) = 2\mathbb{E}\left[\binom{n}{2}\right] - \mathbb{E}[X] + \mathbb{E}[X]^2 = 2\left(\frac{1}{2}\right) - 1 + 1^2 = 1$$

EXAMPLE

Coupon collector problem

n types of coupons. Each time, get unif rand coupon, indep of prev coupons.

Goal: Collect all coupons

$X = \#$ coupons collected before we get every type

$X_i = \#$ addtl coupons needed to go from having $i-1$ types to having i types

$$X_i \sim \text{Geom}\left(\frac{n-i+1}{n}\right)$$

$$\text{Then, } X = \sum_{i=1}^n X_i \Rightarrow \mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n \frac{n}{n-i+1} = n \sum_{i=1}^n \frac{1}{i} \approx n \ln n$$

$Y = \#$ types of coupons we have a unique copy of when we complete our first set

Example: $n=5$, 3 1 4 1 5 2 $\Rightarrow X=6, Y=4$

We relabel coupons based on the order they appear

Define $A_i = \{\text{coupon } i \text{ appears uniquely}\}$

Define $I_i = \mathbb{I}_{A_i}$, then $Y = \sum_{i=1}^n I_i \Rightarrow \mathbb{E}[Y] = \sum_{i=1}^n \mathbb{P}(A_i)$

In other words, $A_i = \{\text{after the first copy of } i, \text{ we get } i+1, i+2, \dots, n \text{ before we get } i \text{ again}\}$
 $= \{\text{among } i, i+1, i+2, \dots, n, i \text{ is the last coupon}\}$

By symmetry/uniformity, any of those coupons is equally likely to be the last
 $\therefore P(A_i) = \frac{1}{n-i+1} \Rightarrow E[Y] = \sum_{i=1}^n P(A_i) \approx \ln n$

Notice, $E\left[\binom{Y}{2}\right] = \sum_{1 \leq i < j \leq n} P(A_i \cap A_j)$

Define $S_{ij} := \{\text{Among } \{i, i+1, \dots, j\}, \text{ coupon } i \text{ appears last}\}$

Then, $P(S_{ij}) = \frac{1}{j-i+1}$

Define $T_{ij} := \{\text{Among } \{i, j, j+1, \dots, n\}, \text{ coupons } i \text{ and } j \text{ are the last 2}\}$

Then, $P(T_{ij}) = \frac{2}{(n-j+2)} \leftarrow i \text{ or } j \text{ last}$
 $\leftarrow \# \text{ pairs that could be the last 2 (unif)}$

As S_{ij}, T_{ij} depend on disjoint sets of coupons, thus they're indep

$\therefore P(A_i \cap A_j) = P(S_{ij} \cap T_{ij}) = P(S_{ij}) P(T_{ij}) = \frac{1}{j-i+1} \cdot \frac{2}{(n-j+2)}$
 $\uparrow$
 indep

$\therefore E\left[\binom{Y}{2}\right] = \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) = \sum_{1 \leq i < j \leq n} \frac{1}{j-i+1} \cdot \frac{2}{(n-j+2)}$

GENERALIZATION

Suppose $X = \sum_{i=1}^n I_i$, $I_i = \mathbb{1}_{A_i} = \begin{cases} 1, & A_i \text{ occurs} \\ 0, & \text{o/w} \end{cases}$

Then, $E\left[\binom{X}{k}\right] = E[\# \text{ k sets of events occurring}] = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k})$